Computer Applications in Business

Lecture 1
Introduction to Computer

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Contents

- What is Computer?
- Workflow of Computer
- Characteristic features of computers
- Types of Computer's User
- Classification Of Computer

Computer

- The word computer comes from the word “compute”, which means, “to calculate”.

- Thereby, a computer is an electronic device that can perform arithmetic operations at high speed.

- A computer is also called a data processor because it can store, process, and retrieve data whenever desired.

- A computer is an electronic device, that processes data and performs tasks through a combination of hardware and software components, enabling it to execute instructions, store and retrieve information, and solve complex problems. It is made up of two parts- hardware and software.

Simplest workflow Of Computer

- Computer Input is called Data.
- Computer Obtained after Processing data, based on user's instructions is called information.
- Raw facts and figures which can be processed using arithmetic and logical operations to obtain information are called data.
- Processes that can be applied to data are of two types:
  a. Arithmetic Operations.
  b. Logical Operations.

Types of Computer's User

Computers for Individual User:
Personal electronic devices designed for individual use. Example: Laptops, desktop computers, and tablets used by people for tasks like browsing, writing, and entertainment.

Computers for Organizations:
Information processing systems tailored for business or institutional needs. Example: Servers, mainframes, and networked computers used by companies for managing data, communications, and operations.

Computers in Society:
The broader impact and integration of computers in everyday life.
Example: Smartphones, ATMs, and online platforms shaping how people communicate, access information, and participate in the digital age.

Classification Of Computers

All modern Computers and computing devices use microprocessors. Microprocessor speeds and storage capacities are skyrocketing day by day.

Computers are classified on the basis of their use or size:

- Desktop
- Laptop
- Tablet
- Server
- Mainframe
- SuperComputer

Desktop:

Personal Computers (PCs) designed for use by an individual at a fixed location.

- IBM was the first company to introduce and popularize use of desktops.
- A Desktop Unit has

✔ CPU (Central Processing Unit)
✔ Monitor
✔ Keyboard
✔ Mouse

Laptop:

- Compact and portable personal computer introduced in 2000s.
- Enough processing power and storage capacity to be used for all office work, website designing, software development and audio/video editing.
- Laptops run using batteries and connect to networks using Wi-Fi chips.

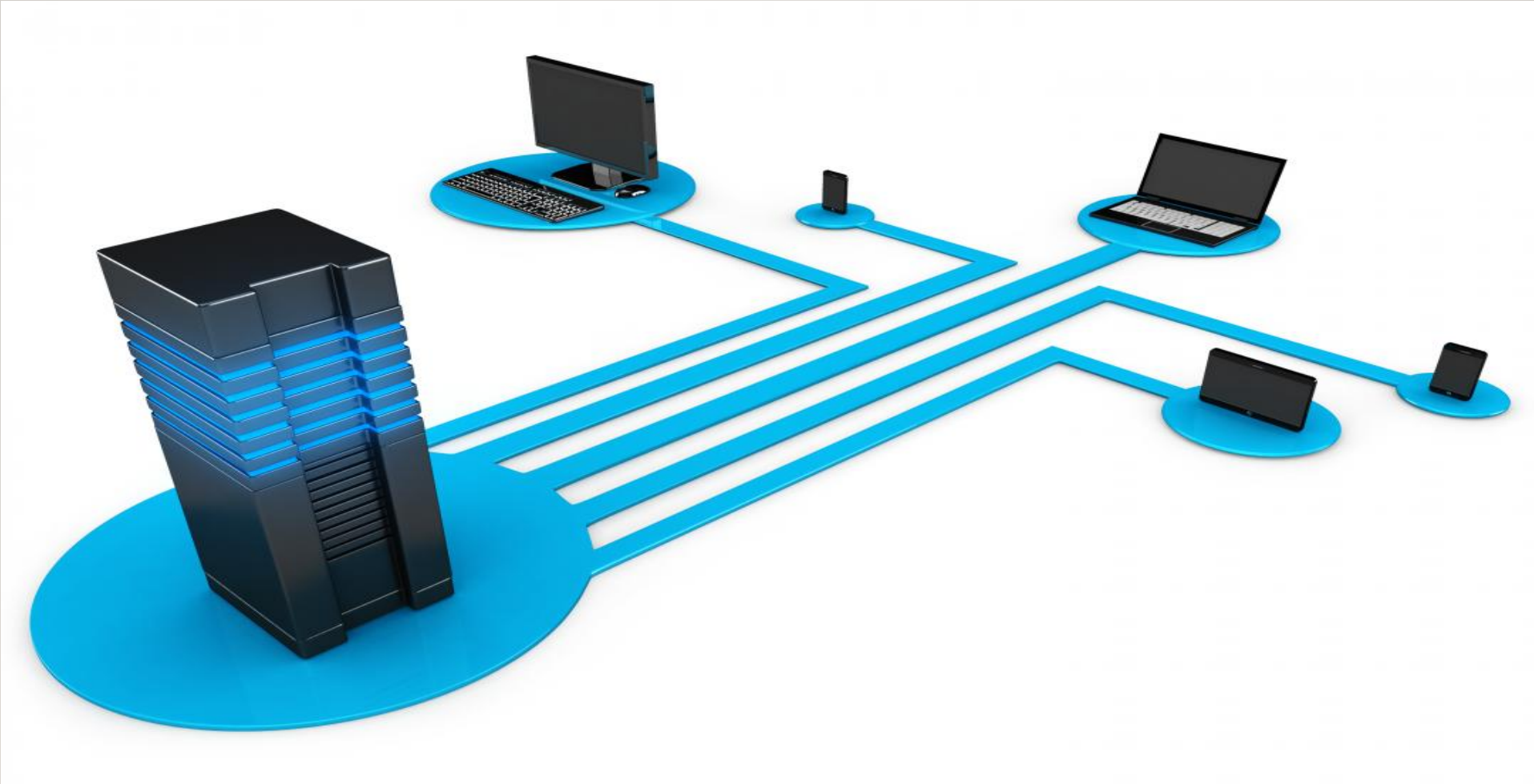
Super Computer:

- Fastest computers on earth
- Used for carrying out complex, fast and time intensive calculations for scientific and engineering applications.
- Speed or performance is measured in teraflops, i.e. 10^12 floating point operations per second.
- Most Common Uses of Supercomputers include:

✔ Molecular Mapping and research
✔ Weather forecasting
✔ Environmental research
✔ Oil and gas exploration

Tablet:

- Processing power of a Desktop.
- Touch sensitive screen of typically 5 to 10 inches.
- Applications that run on tablets are called apps.
- Operating system s provided by Microsoft or Google.



Server

Server:

- **High processing speeds.**
- **Provides one or more services to other systems on the network.**
- **Can handle multiple requests simultaneously.**
- **Most commonly found servers on Networks include.**

❖ **File or storage server**
❖ **Game Server**
❖ **Application Server**
❖ **Database Server**
❖ **Print server**

Main Frame:

- Important features of mainframes are:

- Big in size.
- Hundred times faster than servers.
- Very expensive.
- In built hardware, Software and firmware security features.